

Step 4 in Proof of Ricardian Proposition

① To solve backward for

$\bar{b}_k$:

$$C_t + \frac{\bar{b}_{t+1}}{R} = \hat{y}_t + \bar{b}_t ;$$

$$\boxed{\hat{y}_t = y_t - z_t}$$

$$t = k-1$$

$$C_{k-1} + \frac{\bar{b}_k}{R} = \hat{y}_{k-1} + \bar{b}_{k-1}$$

$$\bar{b}_k = R (\hat{y}_{k-1} - C_{k-1} + \bar{b}_{k-1})$$

$$\Rightarrow \bar{b}_{k-1} = R (\hat{y}_{k-2} - C_{k-2} + \bar{b}_{k-2})$$

$$\begin{aligned} \bar{b}_k &= R (\hat{y}_{k-1} - C_{k-1}) \\ &\quad + R^2 (\hat{y}_{k-2} - C_{k-2}) \\ &\quad + R^2 \bar{b}_{k-2} \end{aligned}$$

Iterate ...

$$b_k = \sum_{j=1}^k R^j (y_{k-j} - (x_{k-j}))$$

$$+ R^k \bar{b}_0$$

$$= \sum_{j=1}^k R^j (\underbrace{y_{k-j} - \tau_{k-j}}_{(x_{k-j})})$$

$$+ R^k \bar{b}_0$$

② Finishing the proof

$$b_k - \bar{b}_k = \sum_{j=1}^k R^j (\bar{\tau}_{k-j} - \tau_{k-j})$$

$$R^{1-k}(b_k - \bar{b}_k) = R R^{-k} \sum_{j=1}^k R^j (\bar{\tau}_{k-j} - \tau_{k-j})$$

$$= R \left\{ R^{1-k} (\bar{\tau}_{k-1} - \tau_{k-1}) \right. \\ \left. + R^{2-k} (\bar{\tau}_{k-2} - \tau_{k-2}) \right. \\ \left. + \dots + R^0 (\bar{\tau}_0 - \tau_0) \right\}$$

$$= R \left\{ \sum_{t=e}^{k-1} R^{-t} (\bar{\tau}_t - \tau_t) \right\}$$

⊖

Then you can
show that $\bar{b}$ satisfies Transversality